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Compound Hydro-Geomorphic Hazards in European Mountain Regions: A Roadmap for Research on Postfire and Frozen Soil Dynamics

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Mountain regions in Europe increasingly face compound runoff-related hazards under a changing climate. This article compares postfire surface runoff (PFSR) and frozen-soil surface runoff (FSSR), 2 processes typically studied separately but sharing key meteorological triggers, as well as initiation and propagation mechanisms. Both are triggered by short, high-intensity rainfall acting on infiltration barriers: hydrophobic layers in burned soil and pore ice in frozen soil. Despite distinct thermal and seasonal contexts, PFSR and FSSR exhibit convergent process chains characterized by reduced interception, limited infiltration, surface ponding, and efficient sediment entrainment. Their occurrence increasingly transcends expected seasonal regimes, reflecting climate-driven shifts in fire activity, snow dynamics, and freeze–thaw cycles. Building on these parallels, this study proposes a research framework and outlines a future research agenda that links mapping, monitoring, experimentation, modeling, and governance. It advocates integrated observation networks combining remote sensing and in situ techniques, controlled experiments to quantify thermohydraulic and erosional processes and changes in vegetation cover, and physical models capturing preferential flow and transient infiltration barriers. For risk management, the article stresses predictive early warning systems that provide sufficient lead time to manage evacuations and infrastructure protection. By identifying shared mechanisms and converging research needs, this contribution supports a future research agenda and cross-disciplinary strategies for mitigating emerging compound hazards in alpine environments.

Keywords: *postfire surface runoff; frozen soil surface runoff; compound hazards; infiltration barriers; debris flows; hydrological extremes; climate change; freeze–thaw processes; monitoring and modeling; risk governance.*

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Introduction

Mountain regions are increasingly exposed to hydrological extremes caused by climate change. These include water-related events such as heavy rain, snowmelt floods, flash floods, and landslides, as well as dry extremes such as droughts and wildfires (IPCC 2023). Two types of compound events have gained prominence due to their challenging nature, increasing relevance under climate change, and comparatively limited investigation in European mountain regions. (1) Postfire surface runoff (PFSR) is triggered when intense rainfall hits fire-altered soils with reduced water infiltration capacity and increased erodibility, generating sediment-laden floods or debris flows (Santi and Rengers 2022; Gartner et al 2024; McGuire et al 2024). (2) Frozen soil surface runoff (FSSR) can occur when seasonally frozen or snow-covered soils inhibit infiltration, leading to surface ponding and overland flow (Bardou and Delaloye 2004; Baselt and Heinze 2021).

Although PFSR and FSSR are typically studied as separate processes and may appear unrelated at first glance, both are controlled by similar boundary conditions. In general, they are not directly linked in space and time, and typically occur in different seasonal contexts. However, they may affect the same regions sequentially, for example, when burned catchments are subsequently exposed to freezing conditions. Their increasing importance highlights the value of analyzing the mechanisms they share within an integrated framework. Monitoring, modeling, and management strategies rarely account for possible shared features or synergies, leaving potential for knowledge transfer and joint countermeasures unused. A critical research gap is the absence of a synthesis that systematically examines these commonalities, with the dual aim of advancing process understanding and informing the development of cross-applicable strategies for impact assessment and risk mitigation.

This article highlights that PFSR and FSSR share predisposing factors and triggers leading to comparable process chains. These similarities call for common experimental, modeling, and governance approaches. We propose an interdisciplinary framework to identify shared mechanisms and compound hazard scenarios in rainy mountain regions characterized by steep and fire- or freeze-prone landscapes, with specific reference to the European context. Such compound scenarios may also include situations where both processes affect the same regions or

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catchments, even if not simultaneously. By synthesizing process types, this contribution aims to support monitoring and management techniques and advance an integrated research agenda for increasingly frequent compound runoff-related hazards (Alcántara-Ayala 2025; Arango-Carmona et al 2025; Brett et al 2025; Melzner et al 2025).

State of the art

The literature considered for this article is mainly derived from recent and ongoing research projects in which the authors have been involved and in which systematic literature research and assessments of postfire and frozen soil processes have been carried out. Building on these foundations, we selected key references representing the most relevant studies for the topics addressed, with specific emphasis on the European context. The review was further complemented by recent publications from 2022–2025 identified through targeted searches in Web of Science, Scopus, and Google Scholar, using keywords such as "wildfire," "postfire surface runoff," "frozen soil hydrology," "compound hazards," "debris flow," "freeze–thaw processes," "preferential flow," and "risk governance" to ensure that the overview and proposed research agenda reflect current developments in monitoring, modeling, and hazard management.

Wildfire, postfire surface runoff, and debris flow

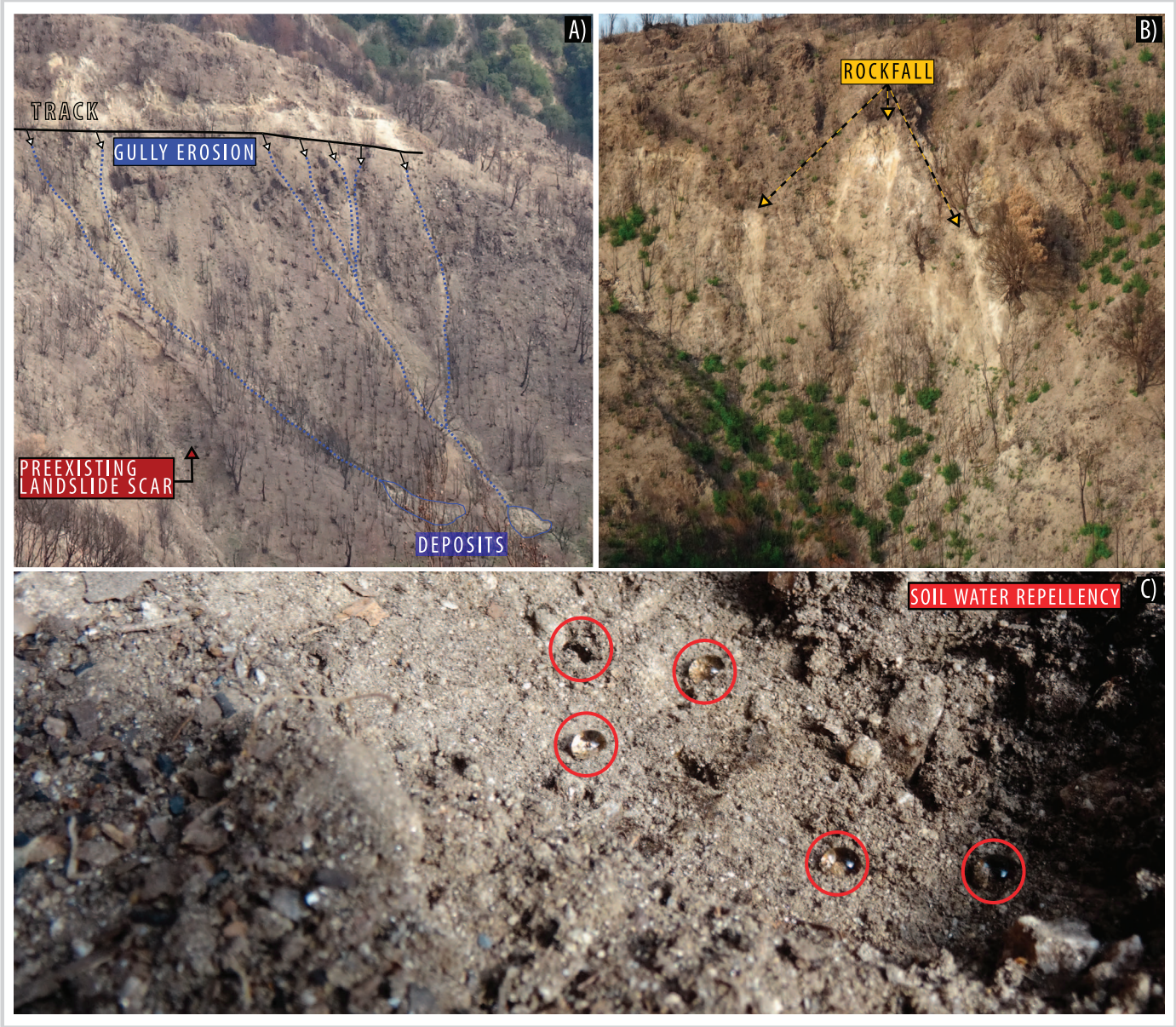
Wildfires have increased globally in frequency and severity, particularly in mountain regions (Müller et al 2020; Liu et al 2024; Selander et al 2025). Europe ranks among the most fire-prone areas worldwide (Grünig et al 2026), due to declining agricultural activity and rising fuel loads (Moreira et al 2011), more frequent droughts and heatwaves (Elia et al 2024), and a sharp increase in fire season length (Wu et al 2015; Bowman et al 2020; Dupuy et al 2020; Sayedi et al 2024). Mountain regions thus present extensive landscapes prone to both wildfires and postfire processes that depend, in turn, on a variety of predisposing factors (Melzner et al 2025). In steep mountain environments, wildfire disturbances often initiate a cascade of hydrological and geomorphic responses. Vegetation loss, soil heating, and ash deposition can reduce infiltration capacity and surface protection cover, promoting rapid surface runoff during intense rainfall. This runoff may subsequently mobilize sediment, trigger landslides, and provide the material for torrential debris flows. PFSR develops when rainfall impacts burned hillslopes, initiating erosion and sometimes mass movements (Figure 1), even on moderate slopes or small scars (eg Cannon and Gartner 2005; Cannon and DeGraff 2008; Abbate et al 2019; Ebel and Moody 2020; Raymond et al 2020; Sarro et al 2021; Gartner et al 2024; Jacobson et al 2024; Zhou et al 2025).

In Europe, intense postfire erosion has been documented along burned hillslopes with reduced soil cohesion, root strength, and hydraulic conductivity (Peduto et al 2022; La Porta et al 2025). Morphometric studies of burned watersheds prone to postfire soil erosion and debris flows show steep relief and short concentration times ($<$10 min), explaining rapid sediment responses (Esposito et al 2019). Fire-prone mountain regions often coincide with catchments that are already predisposed to rapid runoff generation and debris flow initiation (Ebel and Moody 2020). Steep slopes, shallow soils, short concentration times, and abundant loose sediment create geomorphic conditions that favor rapid hydrological response to intense rainfall. When a wildfire removes vegetation and alters soil hydraulic properties, these preexisting factors can significantly amplify postfire runoff and debris flow hazards. Generally, the sediment cascade is driven by wildfires that alter soil properties controlling infiltration and runoff: hydrophobic layers (~5 cm) due to ash or fine particle sealing (Tang et al 2019), hyperdry conditions (Moody and Ebel 2012), and mineralogical changes (Esposito et al 2017) may contribute to reduced infiltration (Gorr et al 2024). Ash can either clog pores and increase runoff (Onda et al 2008) or, depending on its properties, promote infiltration and reduce splash erosion (Woods and Balfour 2008; Melzner et al 2019). When hillslopes are impacted by intense rainfall shortly after or in the months following wildfires, runoff-driven erosion progressively entrains sediment, with flows often bulking into hyperconcentrated flows or debris flows downstream (Jordan 2016; Santi and Rengers 2022; Gorr et al 2024). Laboratory tests simulating burned soil conditions show that the combustion of organic matter can reduce the yield stress of fire-burned debris mixtures by up to 100% and viscosity by up to 80%, increasing debris flow mobility and providing critical rheological parameters for hazard models (Gao et al 2025). While triggering rainfall is usually short, subhourly intensities are often decisive. In southern Italy, the peak intensity of triggering bursts within 10 and 30 min ranges between 30–138 mm/h and 19–105 mm/h, respectively, with modest return periods (1–3 years) showing that such events are not extreme (Esposito et al 2023).

While postfire process cascades are well known and well studied in the United States, the Mediterranean region, and Australia, systematic documentation of postfire geomorphic processes remains limited in the European Alps (Melzner et al 2025). Inventories of slope instabilities vary in completeness and thematic focus, directly influencing hazard and risk assessments (Guzzetti et al 2012; Steger et al 2016, 2017; Melzner et al 2020, 2023). In situ observations are equally challenging, as postfire debris flows can develop within minutes, leaving only a narrow time window for installing measurement systems after a fire (Kean et al 2011) and hampering long-term monitoring. Empirical studies have examined soil heating and water repellency (DeBano 2000), but temporal dynamics of infiltration capacity, repellency, and sediment availability remain poorly constrained, limiting model calibration and validation (Lopes et al 2021). Numerical models range from simplified kinematic wave or Green–Ampt schemes that simulate runoff generation and debris flow timing with few parameters (McGuire et al 2018; Rengers et al 2019) to fully coupled morphodynamic frameworks linking runoff, entrainment, and deposition with probabilistic surrogate models for sensitivity and uncertainty analysis (Spiller et al 2025). However, parameter transferability, spatial heterogeneity, and the integration of long-term recovery trajectories into predictive frameworks remain key challenges (Lopes et al 2021).

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FIGURE 1 Photographs collected in Calabria (Italy) after a wildfire on 26 July 2025 and 2 severe rainstorms in mid-August: (A, B) postfire geomorphic processes, (C) water drops (circles) on a strongly hydrophobic soil. (Photos by Giuseppe Esposito).



Erosive surface runoff over frozen soils

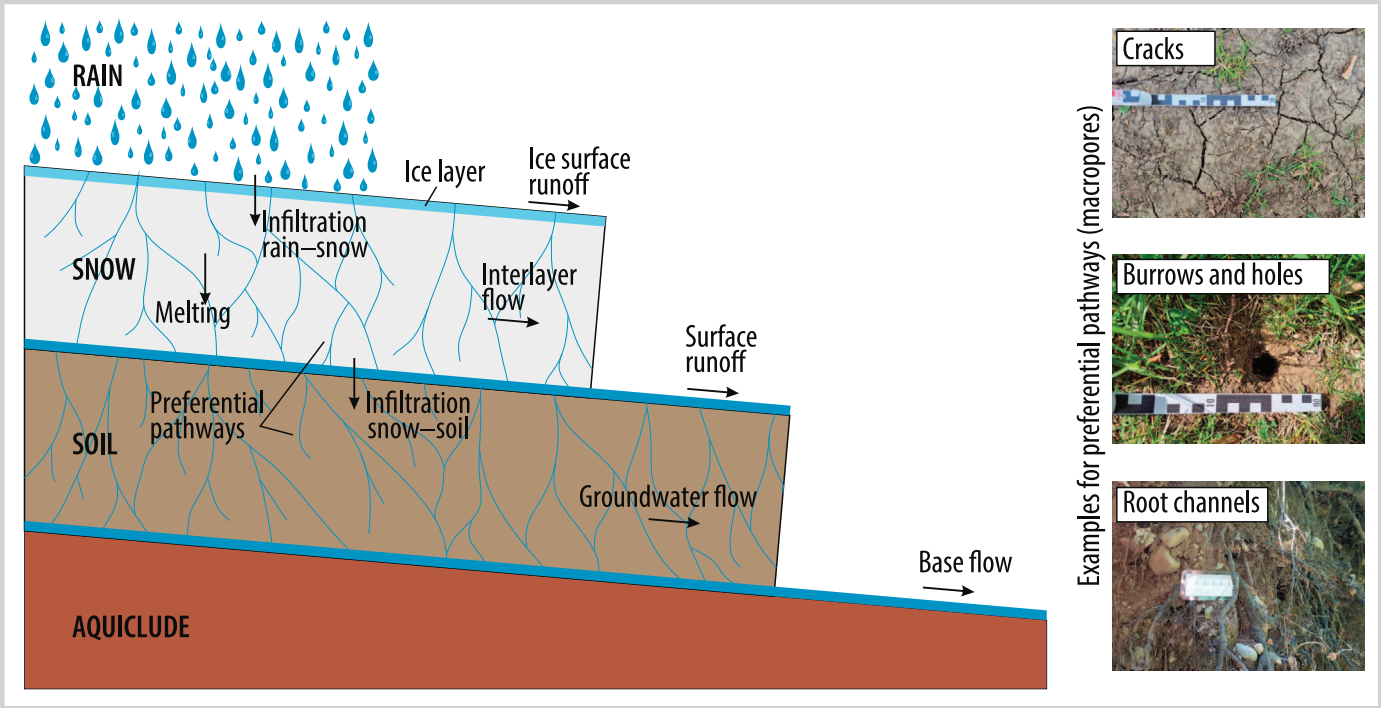
When rain falls on frozen soil, infiltration is often sharply reduced due to pore blockage by ice (Burt and Williams 1976; Kane and Stein 1983; Iwata et al 2011). Consequently, precipitation or snowmelt contributes directly to FSSR (Figure 2), which can rapidly erode sediment and trigger slope failure. These effects are particularly relevant during transitional seasons, when freeze–thaw cycles and intermittent snow cover alter surface hydrology (Bardou and Delaloye 2004; Ala-Aho et al 2021).

The dominant hydrological response in a catchment depends on thermal-hydraulic conditions, antecedent soil moisture, and structural soil properties such as porosity, hydraulic conductivity, and pore connectivity (Baselt and Heinze 2021). In seasonally frozen soils, the frost-affected zone is typically limited to the upper 40 cm, where subzero temperatures prevail, while deeper soil layers remain above 0°C and gradually warm with depth to 8–10°C (Iwata et al 2011; Ala-Aho et al 2021). This vertical temperature gradient, governed by surface energy exchange and thermal inertia, creates a transient freeze–thaw interface that controls infiltration during winter and early spring precipitation (Demand et al 2019; Stuurop et al 2022). Open macropores such as cracks, root channels, or wormholes can disrupt this interface, allowing air, vapor, and water to bypass the frozen matrix (Stähli and Stadler 1997; Beven and Germann 2013).

Vertical soil column experiments have demonstrated that, under freezing conditions, infiltration is restricted to preferential pathways along unfrozen macropores, while matrix flow is significantly impeded (Stadler et al 2000; Weigert and Schmidt 2005; Watanabe and Kugisaki 2017; Grant et al 2019; Holten et al 2019; Pittman et al 2020). Recent laboratory and field experiments simulate slope-scale processes under controlled temperature and

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FIGURE 2 Schematic of infiltration on frozen soil: rainfall on snow and frozen layers limits infiltration and increases runoff (extended from Baselt and Heinze 2021).



precipitation regimes representative of mountain environments (Mohammed et al 2019; Rey et al 2021; Bauer et al 2026). These experiments highlight that initial soil water content controls infiltration depth and thaw rates, while macropores govern the onset and timing of FSSR. At the same time, system responses are highly sensitive to precipitation intensity, geology, soil structure, air temperature, and the duration of freezing.

Field observations confirm the hydrological significance of frozen soils. During rain events, infiltration occurs rapidly and unevenly through preferential pathways (Conway and Benedict 1994), while basal ice layers can impede infiltration and trigger FSSR (Stähli et al 2004). Snowpacks act as dynamic vadose zone extensions where preferential flow governs infiltration timing (Webb et al 2018). Although macropore-driven infiltration under frozen conditions is documented (Stähli et al 1999; Webb et al 2018; Fang et al 2024), boundary conditions, such as antecedent moisture, freezing depth, and rainfall intensity, remain uncertain, and spatial macropore heterogeneity complicates understanding of thermohydraulic processes.

Although coupled thermohydraulic processes are hydrogeologically important, most physical models still assume homogeneous pore structures and local thermal equilibrium, overlooking preferential macropore flow. Hence, their applicability to seasonally frozen soils and thaw-induced slope failure is limited (Heinze 2021). Existing freeze–thaw and permafrost schemes rarely resolve preferential flow under partially frozen conditions (Mohammed et al 2018). Richards-based approaches further assume matrix-dominated infiltration and thermal equilibrium, causing errors under steep thermal gradients or heterogeneous flow (Hansson et al 2004; Kurylyk et al 2014; Grenier et al 2018). Recent dual-permeability and local thermal nonequilibrium (LTNE) models (Larsbo et al 2019; Heinze and Blöcher 2019; Heinze 2025; Khanahmadi et al 2026) improve realism but remain weakly validated.

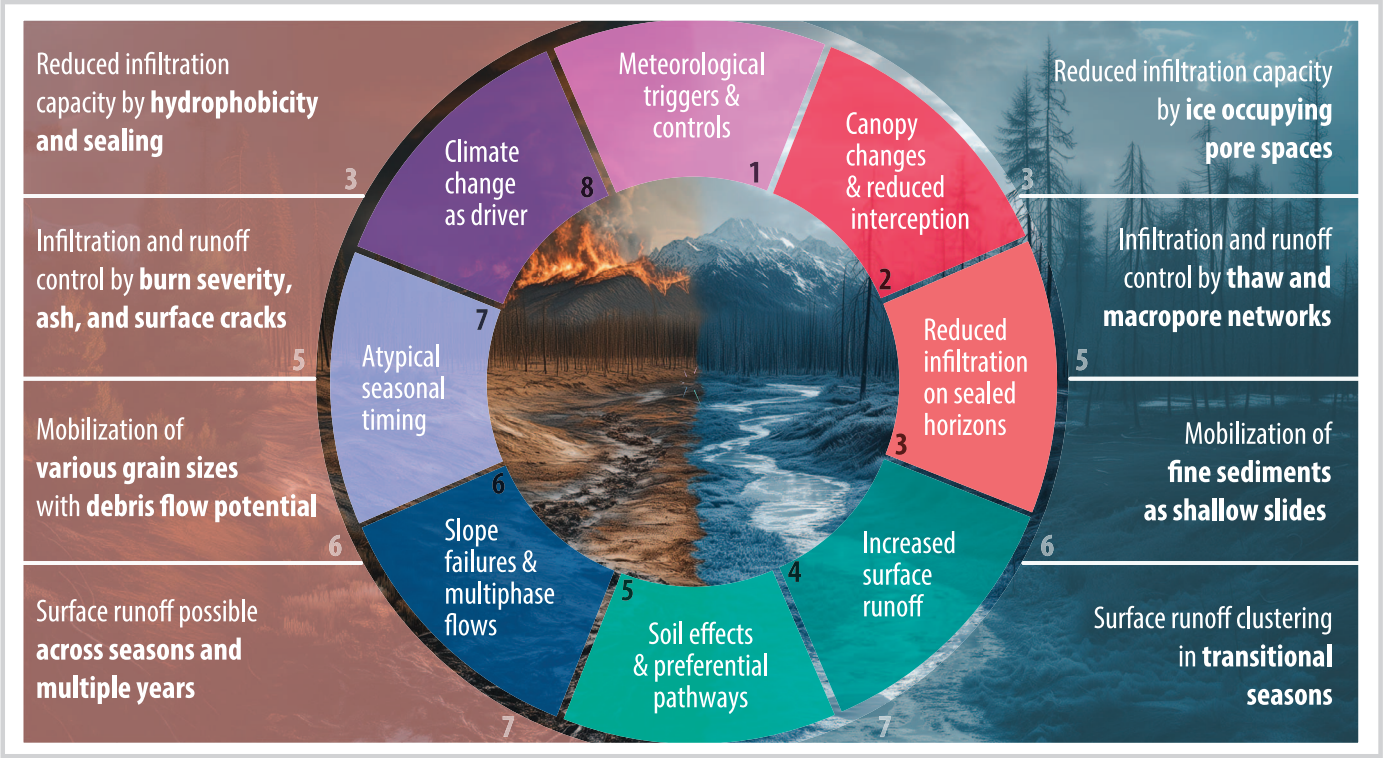
Process similarities and divergences

PFSR and FSSR occur in different thermal and seasonal contexts and are not directly related. However, they share key predisposing factors and mechanisms, 8 of which are introduced below (Figure 3).

1. **Meteorological triggers** exert strong control over the often abrupt initiation of PFSR and FSSR. Short, high-intensity rainfall or rapid transient thaw events can initiate sudden runoff. Limited storage capacity and short concentration times promote flow responses that may lead to sudden erosion and sediment entrainment in both systems (Haslinger et al 2025).
2. **Interception** is reduced in both systems. After wildfires, canopy and litter removal expose the soil to precipitation, while, in frozen soil environments, deciduous canopies are seasonally absent. In both cases, a larger fraction of rainfall reaches the ground, increasing the kinetic energy imparted to the surface.
3. **A near-surface, low-permeability horizon** limits infiltration. It is typically about 5–10 cm deep. The nature of this infiltration barrier differs between systems. In burned catchments, hydrophobic layers may form or be enhanced as chemically mediated sealing through combustion-induced changes in organic or mineral compounds (Doerr et al 2000). In frozen soil, ice within pore spaces creates a low-permeability barrier.
4. **Surface water ponding above these infiltration barriers**, in combination with steep terrain and reduced surface roughness, increases surface runoff. Usually, these early

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FIGURE 3 Schematic of shared mechanisms (circle) and underlying process differences (side panels) for both postfire (left panel) and frozen soil (right panel) environments (background generated with AI, Midjourney, 2025).



flow pulses of PFSR and FSSR are dominated by rapid overland flow.

5. **Soil and rock properties, and preferential pathways** such as cracks and other macropores influence infiltration and runoff in both systems. In postfire settings, ash and char variably affect infiltration, depending on burn severity and ash composition. Observations range from pore clogging and surface crusting that inhibit infiltration (Onda et al 2008) to temporary water retention by black char, which can prevent sealing and reduce splash erosion (Woods and Balfour 2008; Wittenberg 2013; Shtober-Zisu et al 2018). Additionally, surface cracks formed during burning may locally enhance percolation. In frozen soils, infiltration is progressively restored during thaw or along macropore networks that transmit water and heat downward, locally accelerating thaw and altering runoff thresholds (Watanabe and Kugisaki 2017; Bauer et al 2026).
6. **Sediment mobilization, sediment transport, and slope failure** are involved in both systems and can lead to multiphase mass flows, often evolving as cascading sediment transfer processes across hillslopes and channel networks (eg Murphy et al 2019). Burned hillslopes supply abundant sediments of various grain sizes through vegetation loss and ash redistribution, often resulting in a wide spectrum of mobilizable material from fine ash to coarse debris. Through rapid PFSR entrainment, these sediments may eventually contribute to debris flow formation (Melzner et al 2025), typically dominated by runoff-driven erosion and entrainment processes. In frozen soil environments, shallow saturation over icy layers and thaw-weakening increase pore pressure and reduce shear strength, whereas thawing surface horizons may detach and move downslope along frozen–unfrozen interfaces, promoting shallow sliding (Bardou and Delaloye 2004) and reflecting different failure kinematics compared to postfire conditions. In cases of larger-scale slope failures, a similarly wide range of sediment grain sizes may be mobilized.
7. **Expected seasonal regimes** increasingly deviate in both processes, with timing and persistence diverging. PFSR is often triggered by the first high-intensity storms following fire and may occur year-round, depending on burn severity, rainfall duration and intensity, and site recovery (Melzner et al 2022). Approximately 87% of postfire debris flow events occur within the first 2 years after a fire. McGuire et al (2024) further show that more than 80% of runoff-generated postfire debris flow events occur in the first year, whereas roughly 82% of landslide-generated events occur more than 1 year after a fire. Wildfire, thus, can reduce protection for months to years (Melzner et al 2019; Esposito et al 2023). In contrast, FSSR tends to cluster in transitional seasons only, with event likelihood depending on snowpack, antecedent moisture, and freeze–thaw cycles (Pittman et al 2020).
8. **Climate change** is expected to increase the magnitude and frequency of PFSR and FSSR. Rising temperatures, longer fire seasons, more frequent heatwaves and droughts, as well as altered snow regimes, shallower seasonal freezing, and more frequent freeze–thaw cycles collectively expand the windows of occurrence and intensify triggering conditions (IPCC 2023).

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Future research agenda

The proposed research agenda builds on the findings and experiences gained from recent and ongoing research projects addressing postfire and frozen soil hydrological processes. These projects provided empirical insights and conceptual advances that form the basis for identifying shared mechanisms, methodological needs, and cross-disciplinary opportunities. Building on this, the following sections outline priority areas for future work in monitoring, experimentation, modeling, and risk governance to improve understanding and management of compound runoff-related hazards in mountain regions.

Monitoring

Improving understanding of runoff-related hazards in mountain environments requires monitoring strategies capable of capturing the dynamics of surface runoff in both postfire and frozen soil settings. Important steps in the development of a monitoring strategy include defining a catalog of field evidence for burn severity indicators related to vegetation, soil, and rock (Melzner et al 2025) and a catalog of methods to determine burn severity. Remote sensing, in situ instrumentation, and geophysical approaches provide complementary perspectives, but each comes with specific advantages, limitations, and quality requirements. Airborne Laser Scanning (ALS), Terrestrial Laser Scanning (TLS), Unmanned Aerial Vehicle Light Detection and Ranging (UAV LiDAR), and photogrammetric surveys are powerful tools to detect sediment sources, deposition zones, and geomorphic and vegetation changes. Their applicability has been demonstrated in postfire contexts (Rengers et al 2016), as well as in frozen soil and snow settings (Deschamps-Berger et al 2025). In the wide spectrum of remote sensing technologies, synthetic aperture radar (SAR) systems are also useful for burn severity mapping (Tanase et al 2015) and for soil moisture and surface roughness investigations to assess postfire hydrological response or snow and frozen ground conditions.

LiDAR penetrates vegetation and records multiple attributes per pulse, whereas photogrammetry reconstructs detailed 3D surfaces from overlapping UAV or terrestrial images (Melzner and Schiller 2022). Technique selection depends on terrain: TLS for steep faces, UAV LiDAR for mixed slopes, and ALS for large-scale coverage. In frozen soil contexts, these tools document snowpack, frost, and microtopographic changes during freeze–thaw cycles (Jadoon et al 2015; Nakata et al 2021; Cho et al 2025). Satellite multispectral data from Landsat and Sentinel complement such high-resolution datasets by providing consistent, long-term regional observations, enabling analyses of burn severity, vegetation recovery, soil moisture, and snow dynamics across mountain ranges (Wulder et al 2022; Lewińska et al 2024; Melzner et al 2025).

Continuous in situ monitoring is essential to observe triggering conditions and event dynamics in real time. In both burned and frozen environments, soil water content (SWC), measured at different depths, will reveal infiltration behavior during rainfall on burned slopes and during rain or melt on frozen soils. In addition to traditional time domain reflectometry (TDR) and frequency domain reflectometry (FDR) probes and capacitance-based or, more generally, dielectric soil moisture sensors (Watanabe and Kugisaki 2017; Bauer et al 2026; Salmabadi et al 2026), novel methods such as cosmic-ray neutron sensing extend the observation scale to several hundred meters and depths up to ~80 cm (Zreda et al 2008). Coupling SWC probes with tensiometers may provide information on matric suction, critical for both shallow slope failures in burned terrain and for thaw weakening in frozen soils (Rengers et al 2020). Seismic monitoring has proven effective in detecting debris flow surges (Porter et al 2023) and could similarly be applied to identify erosive runoff pulses in frozen soil settings. Optical and thermal cameras, pressure transducers in channels, and stage gauges offer further opportunities to characterize the timing, type, and magnitude of runoff processes (Kean et al 2011; Dematteis et al 2025).

Data quality must be defined through method-specific criteria. For LiDAR, these include point density, coverage, classification accuracy, and acquisition timing (Melzner and Schiller 2022). In photogrammetry, positional and shape accuracy and image resolution determine the detection of fine-scale features such as macroscopic soil sealing, ash layers, or thaw-induced cracks (Melzner et al 2019; Melzner and Schiller 2020). High-resolution data enhance the capture of small-scale variability in burned and frozen soils but increase computational effort. Integrating multiscale remote sensing with continuous in situ monitoring helps to resolve spatial heterogeneity and temporal dynamics. The main challenge is to develop efficient, scalable, and affordable strategies that yield long-term, high-quality datasets to advance physical models and early warning systems for catchments affected by both fire and frost.

Experimental studies

Future advances in understanding runoff-related hazards will depend on targeted and collaborative experimental approaches, combining controlled laboratory studies with advanced field measurement stations to capture process dynamics across scales. Such investigations are essential for disentangling coupled hydrological and thermal processes, for systematically testing hypotheses of infiltration and runoff generation, and for generating benchmark datasets to calibrate and validate numerical models. In this context, plot- to hillslope-scale rainfall simulation experiments (eg Lechner et al 2026) provide a critical bridge between controlled laboratory conditions and catchment-scale responses, as they enable the quantification of scale-dependent runoff generation processes under realistic boundary conditions. A primary avenue of research concerns the hydrological implications of water repellency and infiltration into hydrophobic soil layers. Initial investigations may employ simple 1-dimensional soil column experiments, extending earlier foundational studies (DeBano et al 1979; DeBano 2000). Subsequent efforts should advance toward 2- and 3-dimensional laboratory setups capable of lateral subsurface flow, evolving heterogeneities, and preferential flow pathways. Designing such systems is technically demanding, as both boundary conditions and evolving variables, including soil moisture, hydraulic conductivity, repellency strength, and pore structure, must be measured with high spatial and temporal resolution. Another promising direction is surface runoff experiments with burned soils and

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ash-rich top layers. These setups could build on lahar-style flume studies (Qin et al 2024, 2025), but should place particular emphasis on the detachment and mobilization of the shallow burned horizon (Movasat and Tomac 2025) and general momentum transfer due to entrainment and erosion (Wetterauer et al 2026). Experiments should include different soil and rock materials, organic residues, and ashes, as well as coarse fire remnants, such as trunks or logs. Woody debris is expected to provide stabilizing structure in some cases and obstruct flow in others. Laboratory studies are uniquely suited to quantify this dual role in promoting or preventing slope instability.

For frozen soil systems, new experimental research is required to resolve key uncertainties in freezing front propagation, macropore behavior, and infiltration–runoff partitioning under freeze–thaw dynamics. The soil specificity of freezing behavior—mediated by grain-size distribution, fine particles, and organic matter—remains poorly constrained. Future laboratory programs should therefore systematically vary soil texture, organic content, and macropore structure to quantify their role in governing freezing progression, infiltration capacity, and runoff initiation. A particularly important frontier is the role of macropores under freezing and refreezing conditions. Existing results suggest that macropores can initially enhance infiltration but subsequently lose continuity once pore ice develops (Bauer et al 2026). The dynamics of these transitions, and their sensitivity to initial moisture content and freeze–thaw sequences, remain insufficiently quantified. Experiments combining rainfall–freeze–thaw cycles, snowmelt simulations, and advanced imaging (eg nuclear magnetic resonance, electrical capacitance tomography) could provide critical process-level insights. Future experiments should also address compound boundary conditions, including repeated rainfall events on frozen soils, rain on snow scenarios, and the combined effects of fire-induced hydrophobicity and subsequent frost. These compound experiments are crucial to reproducing realistic mountain settings where burned soil may later be exposed to seasonal freezing, or where thawing coincides with rainfall.

Modeling integration

Across postfire and frozen soil settings, laboratory experiments and field surveys provide essential testbeds for next-generation physical models. Particularly promising are dual-domain and local thermal nonequilibrium approaches, which explicitly represent preferential pathways, heat transfer, and evolving pore connectivity (Heinze 2021; Khanahmadi et al 2026). Beyond traditional single-domain hydrological frameworks, recent advances in multiphase flow modeling now incorporate thermal processes, ice–water phase transitions, and the resulting feedback on frictional resistance and mobility (Pudasaini and Krautblatter 2014; Feng et al 2023; Munch et al 2024; Pudasaini et al 2024). These developments are critical for hazard assessment, as they enable the simulation of complex rheological behavior of flows enriched with snow, ice, or ash. Nevertheless, benchmark datasets derived from controlled experiments and field surveys are indispensable for systematic model calibration, validation, and parameterization (Lopes et al 2021). The Bussoleno case study in the northwestern Italian Alps illustrates the added value of coupling susceptibility-based triggering analysis with physical runout models (La Porta et al 2025). This underlines that physical models need to account not only for runout but also for temporally and spatially distributed triggering conditions linked to vegetation loss, infiltration reduction, and altered soil strength (Gorr et al 2024).

Further improvements in physical models are required to incorporate mechanical aspects relevant for hazard planning, protection structures, and soft measures, such as bioengineering. A central challenge lies in developing fully coupled hydrological–geomorphic–hydraulic catchment models that account for temporal and spatial changes in land cover, porosity, permeability, and temperature to realistically simulate infiltration, groundwater recharge, and runoff. Findings from hydrogeology underline the necessity of linking runoff, groundwater, and water quality to secure sustainable drinking water supply in mountain regions.

Risk governance

Effective management of postfire and frost-related hazards requires timely and coordinated measures to reduce risks for downstream urban settlements. Population safety should rely on multilayered warning systems that combine local weather alerts with advanced forecasting and monitoring tools. As demonstrated by fatal postfire debris flows in the United States (Kean et al 2019) and Italy (Esposito and Gariano 2025), or by rapid runoff events on frozen ground in Switzerland and Japan (Bardou and Delaloye 2004; Iwata et al 2011), local weather alerts can be insufficient to warn the population efficiently, and more action is thus required. Early warning systems based solely on the exceedance of rainfall thresholds can also be ineffective, since exceedance may occur only minutes before slope or runoff processes are triggered, providing insufficient lead time to manage evacuations or road closures. Early warning protocols should therefore be based not only on rainfall threshold exceedance but also on predictive models that incorporate lead time. In this context, impact-based warning (IbW) approaches are increasingly promoted, shifting the focus from forecasting hazard variables to communicating expected impacts on people, infrastructure, and services. Such frameworks aim to translate forecasts into actionable information by linking hazard intensity with exposure and vulnerability (Potter et al 2018, 2021; Rözer et al 2021). However, most existing IbW systems are focused on single hazards without considering cascading and compound effects, highlighting the need for the development of multihazard approaches in mountain environments (Lumbroso et al 2025).

Weather radar should be systematically employed to monitor rainfall conditions both upwind of vulnerable areas and in locations not covered by rain gauges. Tracking and forecasting the spatial and temporal evolution of convective systems capable of producing intense rainfall must therefore be prioritized. This applies equally to postfire hillslopes with reduced infiltration capacity and to frozen soil catchments where impermeable near-surface ice layers favor rapid runoff generation. Rainfall forecasting models should be coupled with radar and in situ observations to improve nowcasting activities and, consequently, emergency response preparation. Many national meteorological services already operationalize key elements such as radar–gauge merging and radar-based nowcasting. However, full

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end-to-end integration is not yet universal, and limitations remain in complex and exposed terrain due to beam blockage, coverage gaps, and challenges under snowfall or rain on snow conditions.

As shown by Rengers et al (2025), research institutions should work tightly with government management agencies to integrate hazard assessment, forecasting, monitoring, and adaptation activities aimed at reducing compound runoff risks to human life. In parallel with nonstructural measures, local structural interventions should be implemented after wildfires or frost-related disturbance to mitigate, for example, the risk of sediment transport and accumulation on roads or buildings. Key priorities include cleaning and maintaining drainage networks at the intersections with roads and built-up areas (eg culverts), constructing check dams or retention basins upstream of areas at risk, and improving the thermal stability and the hydraulic design of drainage structures exposed to freezing–thawing cycles. In addition, future research should more explicitly address the effects of PFSR and FSSR on vegetation recovery, biodiversity, and soil development. These factors may strongly influence the persistence and evolution of runoff processes, for example, through changes in vegetation cover and associated infiltration and erosion dynamics. Finally, both postfire and frozen soil risk governance demand cross-sectoral coordination between meteorological services, environmental agencies, and civil protection authorities. Joint monitoring and data-sharing platforms are essential for anticipating compound events, which increasingly blur traditional seasonal distinctions and challenge conventional hazard management frameworks in alpine environments.

Implications and conclusions

This article highlights both the convergences and divergences of runoff generation in postfire and frozen soil environments. Both systems exhibit reduced interception, constrained infiltration, rapid surface runoff, and efficient sediment mobilization, underscoring the value of cross-fertilization in monitoring, experimentation, and modeling. At the same time, divergences in the origin of infiltration barriers, seasonal timing, sediment grain size, and failure kinematics highlight a need for caution in using simple analogies and point to the need for process-specific parameterizations. Beyond direct impacts on infrastructure and human safety, these processes can also affect vegetation cover and recovery, soil development, and sediment dynamics in aquatic environments, thereby influencing ecosystem functions in mountain regions.

Future progress will rely on systematically addressing the open questions that arise from the convergences and divergences identified in this article. Building on these shared mechanisms and contrasting features, we have outlined a comprehensive research agenda that spans monitoring, experimental investigations, modeling integration, and risk governance. This agenda highlights that advancing process understanding and hazard management in postfire and frozen soil systems in the high mountains of the Alps is possible only through coordinated and interdisciplinary effort, as shared physical mechanisms enable knowledge exchange and joint methodological development across domains.

Advancing this agenda ultimately depends on collaborative and interdisciplinary research frameworks. Future experimental studies demand joint expertise from cryosphere science, geomorphology, wildfire research, and alpine natural hazard studies, combined with soil physics, hydrology, and engineering to design large-scale, multiprocess laboratory frameworks. Modeling progress requires equally integrative approaches, where empirical benchmark datasets are embedded in international collaborations. Likewise, monitoring strategies must combine remote sensing, in situ instrumentation, and geophysical methods within multi-institutional networks to ensure robust long-term datasets across mountain ranges. Finally, risk governance calls for cooperation beyond science, linking geoscientists, engineers, practitioners, and policymakers to develop effective early warning systems and to implement sustainable, nature-based protection strategies. Taken together, these collaborative efforts are not only critical for advancing process understanding but also for aligning hazard research with the values of sustainable mountain development.

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OPEN PEER REVIEW

This article was reviewed by Jan Klimeš and Matthias Schlögl. The peer review process for all MountainAgenda articles is open. In shaping target knowledge, values are explicitly at stake. The open review process offers authors and reviewers the opportunity to engage in a discussion about these values.

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